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Prenatal Steroids and Metabolic Dysfunction: Lessons from Sheep

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Abstract

Prenatal exposure to excess steroids or steroid mimics can disrupt the normal developmental trajectory of organ systems, culminating in adult disease. The metabolic system is particularly susceptible to the deleterious effects of prenatal steroid excess. Studies in sheep demonstrate that prenatal exposure to excess native steroids or endocrine-disrupting chemicals with steroidogenic activity, such as bisphenol A, results in postnatal development of numerous cardiometabolic perturbations, including insulin resistance, increased adiposity, altered adipocyte size and distribution, and hypertension. The similarities in the phenotypic outcomes programmed by these different prenatal insults suggest that common mechanisms may be involved, and these may include hormonal imbalances (e.g., hyperandrogenism and hyperinsulinemia), oxidative stress, inflammation, lipotoxicity, and epigenetic alterations. Animal models, including the sheep, provide mechanistic insight into the metabolic repercussions associated with prenatal steroid exposure and represent valuable research tools in understanding human health and disease. Focusing on the sheep model, this review summarizes the cardiometabolic perturbations programmed by prenatal exposure to different native steroids and steroid mimics and discusses the potential mechanisms underlying the development of adverse outcomes.

Keywords

cardiometabolic disease; developmental programming; sheep; steroid hormones

INTRODUCTION

The developing fetus is particularly vulnerable to changes in the intrauterine environment during prenatal life. Although a few of these changes may be favorable for the healthy development and survival of the fetus, most of the physiological adaptations occurring during prenatal development in response to insults are detrimental and contribute to the development of adverse health outcomes postnatally. A myriad of animal and epidemiological studies have demonstrated that maternal exposure to adverse conditions

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during gestation owing to nutritional deficits/excess, disease conditions, stress, drug abuse, and/or environmental exposure to endocrine-disrupting chemicals (EDCs) can alter the uterine environment, culminating in the development of adult disease in the offspring (1–4). Therefore, improper lifestyle choices made by the mother during pregnancy pose a threat to the normal development of the fetus (Figure 1).

Barker's (5) developmental origins of health and disease hypothesis has gained considerable attention during the past decades, especially after epidemiological data from the 1944–1945 Dutch famine demonstrated that gestational malnutrition is associated with a significant increase in the incidence of cardiovascular and metabolic diseases in the offspring (6). These findings, corroborated by subsequent preclinical and clinical studies (5, 7, 8), indicate that the prenatal period, a period in which organogenesis and tissue differentiation occur through a tightly controlled and timed process, is a critical susceptibility window for programming the offspring's phenotype. More recently, it has become clear that the mechanisms underlying the fetal origins of adult disease involve reprogramming of the epigenome by environmental factors (9, 10). This reprogramming is possible during prenatal life owing to the plasticity that allows the developing fetus to adopt a phenotype that best suits the environment.

In the context of reprogramming, steroid hormones are major players, as they are involved in modulating normal cell differentiation and organ system development during fetal life (11–13). As such, improper exposure to steroids or steroid mimics during critical periods of development can disrupt the normal developmental trajectory of organ systems, culminating in adult disease. The reproductive and metabolic systems are particularly susceptible to the deleterious effects of steroid excess. The growing fetus can be exposed to inappropriate levels of steroid hormones owing to disease conditions, such as congenital adrenal hyperplasia and polycystic ovary syndrome (PCOS); continued maternal use of contraceptive steroids; maternal use of anabolic steroids; and environmental exposure to EDCs with steroidogenic activity, such as bisphenol A (BPA), diethylstilbestrol, and trenbolone (13–16). The recent increased incidence of estrogen-sensitive cancers, endometriosis, male genital abnormalities, reduced semen quality, and precocious puberty in girls illustrates the looming problem (17, 18).

Animal models provide unparalleled resources to study the effects of perinatal exposure to steroid excess. Most studies have used mice, rats, sheep, and nonhuman primates as research models, and comparative aspects of these different models have been reviewed previously (19, 20). This review focuses primarily on research performed in the sheep, a model in which longitudinal studies focusing on multiple developmental time points have been carried out extensively. There are several advantages of using sheep for endocrine research. Sheep are amenable to different experimental procedures and interventions; their large size allows the performance of detailed and repetitive hormonal profiling and in vivo sampling and measurement of hypothalamic neuropeptides. Because sheep are domesticated, they are kept in a natural environment and not susceptible to the stress effects associated with caging. Moreover, as a precocial species, the developmental trajectory of several organs, such as brain and pancreas, follows a similar pattern as in humans (21) and hence is of translational relevance.

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EFFECTS OF PRENATAL STEROID EXCESS ON CARDIOMETABOLIC FUNCTION

Effects of Prenatal Testosterone Excess

It has long been known that exposure to excess testosterone (T) during fetal development results in phenotypic virilization and neurobehavioral masculinization in female sheep (22). Prenatal T excess has also been shown to induce reproductive and cardiometabolic perturbations in female sheep that recapitulate those seen in women with PCOS (21), an infertility disorder affecting 1 in 10 women of childbearing age (23). Therefore, prenatal T-treated sheep serve not only as an important animal model to study the developmental progression of the PCOS phenotype but also as a testing ground to investigate the efficacy of intervention strategies for treatment and prevention of this syndrome (21).

The use of different experimental paradigms to investigate the effects of prenatal androgens on the health of the offspring has allowed us to gain valuable insight into the susceptibility windows and mechanisms involved in this process. For instance, comparison of sheep treated with the aromatizable androgen T from gestational day (GD) 30 to 90 with those treated from GD 60 to 90 has identified critical periods in which the reproductive and metabolic systems are more sensitive to the programming effects of T. Additionally, comparison of prenatal treatment with T, dihydrotestosterone (DHT; a nonaromatizable androgen), or cotreatment with T and flutamide, an androgen antagonist, has shed some light on the pathways and mechanisms responsible for programming adverse health outcomes. For the purpose of this review, the main effects of prenatal overexposure to T on cardiometabolic outcomes are discussed below. For information on the effects of prenatal T excess on reproductive neuroendocrine and ovarian perturbations in this model, readers are referred to work by Cardoso et al. (18) and Padmanabhan et al. (3).

Fetomaternal endocrine and metabolic milieus and fetal development

To understand the potential mechanisms linking prenatal T excess with development of cardiometabolic dysfunction, it is important to recognize the effects on the fetomaternal endocrine and metabolic milieus. Studies in sheep allow collection of maternal and fetal blood samples and tissues to investigate endocrine, molecular, and cellular alterations at specific phases of fetal growth. These studies have shown that gestational treatment with T increases not only the maternal but also the fetal levels of androgens and estradiol in the female sheep (24, 25). Importantly, the fetal concentrations of androgens achieved with gestational T treatment are similar to the physiological ranges observed in the male offspring (24, 25). Of note, serum concentrations of T during the second trimester are within the male fetus range in ~40% of human female fetuses (27). Gestational T excess has also been shown to reduce maternal progesterone levels, increase circulating concentrations of insulin, decrease medium chain acylcarnitines levels, and disrupt the maternal-fetal correlations for several metabolites (25). Because steroid hormones play an important role in differentiation of organ systems involved in controlling metabolic function (3, 28–30), improper developmental exposure to steroids likely mediates the programming of adult metabolic disorders in prenatal T-treated sheep.

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Another important endocrine alteration seen in prenatal T-treated sheep that likely contributes to the development of adult cardiometabolic perturbations is a reduced bioavailability of several growth factors, such as activin B and insulin-like growth factor 1 (IGF1) (24, 31). The diminished concentrations of bioavailable IGF1 result from two factors: a downregulation of *IGF1* mRNA expression in the liver and an increased abundance of its binding protein (31). These gestational alterations in bioavailability of growth factors are likely to have an impact on fetal organ development and growth. Indeed, gestational T excess results in late-gestation intrauterine growth restriction (IUGR) (24, 32); low birth weight in female offspring (33); and a proportional reduction in the weight of several organs, such as liver, kidney, and spleen (32). Collectively, these findings demonstrate that gestational T excess disrupts the fetomaternal steroidal and metabolic milieus, thus disturbing the overall development of the fetus. The early postnatal catch-up growth associated with the IUGR in this model is also a risk factor for the development of cardiometabolic perturbations (33).

Placental insufficiency is often one of the primary causes of IUGR in response to prenatal insults (34, 35). Consistent with this premise, findings in the ovine model demonstrate that gestational T treatment advanced placental differentiation to maintain placental efficiency early during fetal life but failed to do so later during pregnancy, when placental efficiency was found to be reduced (36). Moreover, studies involving cotreatment with the androgen antagonist flutamide or the insulin sensitizer rosiglitazone indicate that these effects are mediated via androgenic actions of T and are not a function of disrupted insulin homeostasis (36). Therefore, these findings suggest that T excess during pregnancy, acting via the androgen receptor, leads to alterations in placental differentiation and function that likely affect the fetal endocrine milieu and growth, culminating in IUGR and low-birth-weight offspring.

Peripheral glucose-insulin homeostasis

Gestational exposure to T excess promotes insulin resistance and compensatory hyperinsulinemia evident as early as five weeks of age (37). Studies comparing the effects of treatment with T or DHT during gestation suggest that the prenatal programming of insulin resistance occurs via androgenic actions of T (38). However, failure of cotreatment with androgen antagonist to prevent T-induced effects (V. Padmanabhan, M. Puttabyatappa & A. Veiga-Lopez, unpublished observations), coupled with the finding that DHT can be metabolized into 3β-diol and act through estrogen receptor β (39), raises the possibility that the effects of T may be facilitated via estrogenic actions owing to aromatization of T to estradiol. Alternatively, development of insulin resistance in the offspring may be programmed via the maternal hyperinsulinemic status induced by gestational T excess (25). In support of this, our recent observations that prenatal cotreatment with rosiglitazone, an insulin sensitizer, prevented the development of insulin resistance in prenatal T-treated females indicate that gestational alterations in insulin levels likely play a role in programming postnatal perturbations in glucose-insulin homeostasis in this ovine model (40). Observations from the GD 60–90 prenatal T-treated model indicate that the window of susceptibility for programming insulin resistance is confined to a relatively short programming window, spanning from 60 to 90 days of gestation (38).

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Of clinical relevance are our recent findings that postnatal excessive weight gain exacerbated the negative effects of prenatal T treatment on insulin sensitivity and hyperinsulinemia in adult female sheep (38). More specifically, postnatal overfeeding increased the insulin:glucose ratio and cumulative insulin response, while it reduced the insulin sensitivity index during a glucose tolerance test in prenatal T-treated sheep. Of even greater translational importance are the observations that the offspring (F2) of overweight females, which were insulin resistant, also manifested perturbations in insulin dynamics, demonstrating the potential for multigenerational transmission of metabolic imbalances induced by prenatal T excess (38).

Our extensive longitudinal studies investigating glucose-insulin homeostasis in female sheep prenatally exposed to T excess have characterized important fluctuations in insulin sensitivity throughout life (40). During infantile (37) and early juvenile development (38), prenatal T-treated sheep demonstrate a significant reduction in insulin sensitivity. However, during peripubertal life, these females exhibit notable improvements in insulin sensitivity, displaying a greater insulin sensitivity index than control animals during an euglycemic-hyperinsulinemic clamp (41). However, in later adult life, prenatal T treatment results in the reestablishment of insulin resistance in these animals (38). Collectively, these longitudinal observations suggest that a period of compensatory adaptation of insulin target tissues to prenatal exposure to T excess occurs between juvenile (prepubertal) and early adulthood in sheep (40). Our most recent observations of moderate insulin resistance in juvenile females followed by normal insulin sensitivity during early adult life corroborate the predicted trajectory of insulin sensitivity in female sheep prenatally exposed to T excess (Figure 2).

Insulin sensitivity: tissue-specific alterations

Studies investigating the impact of gestational T treatment on the overall expression and activation (phosphorylation) of members of the insulin signaling pathway in metabolic tissues revealed tissue-specific regulation of insulin sensitivity in sheep (42, 43). At the level of the liver, a downregulation of several members of the insulin signaling pathway was observed, consistent with hepatic insulin resistance (43). Additionally, prenatal T treatment prevented the insulin-stimulated phosphorylation of protein kinase B (AKT), a key member of the insulin-signaling cascade, in the liver of adult females (42). Similarly, prenatal exposure to T excess reduced the expression of members of the mitogen-activated protein kinase/extracellular signal-regulated kinase pathway (43) and diminished the phosphorylation of AKT after in vitro insulin stimulation in the skeletal muscle (42). In contrast, prenatal T treatment upregulated several members of the insulin signaling pathway (43) and did not prevent an increase in AKT phosphorylation in the adipose tissue (42), suggesting normal insulin sensitivity. Collectively, these observations point to insulin resistance at the liver and muscle level, while the adipose tissue remains insulin sensitive in prenatal T-treated sheep, an animal model of PCOS phenotype. Similar findings of tissue-specific regulation of insulin sensitivity have been reported in women with PCOS (44–46).

Studies using fetal tissues have demonstrated that tissue-specific alterations in insulin sensitivity are established early in development (42). Interestingly, gestational T treatment increases the phosphorylation of insulin signaling molecules in the liver, suggesting

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increased insulin sensitivity during fetal development (42). Although the exact mechanisms remain unknown, the observations of enhanced insulin sensitivity in the fetal liver in contrast to reduced sensitivity during adulthood suggest that compensatory mechanisms are in place to prevent the development of pathology during early life, though they fail to do so during adult life (42). However, insulin sensitivity in the skeletal muscle was already reduced during fetal life in prenatal T-treated females, which supports a different developmental trajectory of perturbations in the various organ systems (42).

To understand the basis for the tissue-specific regulation, we recently investigated the effects of prenatal T treatment on negative (low-grade inflammation, oxidative stress, and lipotoxicity) and positive (adiponectin and antioxidants) mediators of insulin sensitivity in metabolic tissues (47). Results showed that hepatic liver resistance is a function of increased oxidative stress (increased 3-nitrotyrosine) and lipotoxicity, whereas insulin resistance at the skeletal muscle level is related to lipotoxicity (47). At the visceral adipose tissue level, the insulin-sensitive status appears to result from an increase in antioxidants that likely overrides the increase in proinflammatory cytokines, macrophage infiltration, and oxidative stress (47). Collectively, these findings suggest that local control of positive and negative regulators of insulin sensitivity, such as oxidative stress and lipotoxicity, may be involved in the tissue-specific regulation of insulin sensitivity in this ovine model.

Hypothalamic neurocircuitries

Research on the neuroendocrine regulation of food intake and metabolism has focused primarily on two key hypothalamic neuronal populations, the neurons expressing the orexigenic agouti-related peptide (AgRP) and neurons expressing the anorexigenic peptide proopiomelanocortin (POMC) (28). Both neuronal populations are located in the arcuate nucleus of the hypothalamus, and together they function to regulate energy homeostasis and glucose-insulin metabolism. Both leptin and insulin control energy metabolism and act on AgRP and POMC neurons by inhibiting and activating them, respectively (28, 48). Dysfunctions within this neuronal circuitry can disrupt energy metabolism and perturb glucose-insulin homeostasis, leading to insulin resistance (49).

Neuroanatomical studies in sheep have found that prenatal T excess increases the number of AgRP but not POMC neurons in the middle division of the arcuate nucleus (48). The observations that similar alterations in the number of AgRP neurons were also triggered by prenatal DHT treatment and were blocked by prenatal cotreatment with T and flutamide (androgen antagonist) suggest that activation of the androgen receptor is required for programming this neuroendocrine alteration (48). Moreover, the density of AgRP-containing fibers in the preoptic area, lateral hypothalamus, paraventricular nucleus, and dorsomedial hypothalamic nucleus was also increased by prenatal treatment with T (48). Collectively, these findings suggest that prenatal T excess, acting via the androgen receptor, promotes alterations in the AgRP neuronal circuitry that likely contribute to the postnatal development of insulin resistance and increased risk of obesity in this sheep model.

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Adipocyte size and free fatty acid profile

To better understand the potential causes of lipotoxicity and insulin resistance in this ovine model, we recently investigated the effects of prenatal T excess on adipocyte morphometry. Interestingly, prenatal exposure to T excess increased the ratio of small:large adipocytes in the visceral and subcutaneous adipose tissues during postpubertal development (41) as well as early adulthood (40). Although the dogma in the field has been that adipocyte hypertrophy is linked to the development of insulin resistance (51, 52), studies examining adipocyte size in obese but otherwise healthy subjects have reported that the small versus large adipocyte proportion is higher in insulin-resistant compared with insulin-sensitive individuals (53, 54). Therefore, insulin resistance in some obese patients may stem, at least in part, from failure of a subset of smaller adipocytes to differentiate into fully mature adipocytes with increased lipid storage capacity (55). Consequently, excess free fatty acids are deposited in nonadipose tissues, such as liver and muscle, leading to lipotoxicity, oxidative stress, and related insulin resistance (56–58). Likewise, prenatal T treatment not only resulted in smaller adipocytes (40, 41) and ectopic lipid accumulation in the liver and skeletal muscle (47) but also increased total and saturated free fatty acids, such as palmitic acid, in female sheep (41). Therefore, these findings suggest that the increase in the relative proportion of small adipocytes may be the earliest event in the development of metabolic dysfunctions, such as dyslipidemia, ectopic lipid accumulation, and insulin resistance in prenatal T-treated sheep.

Cardiovascular function

In addition to metabolic perturbations, studies in sheep have provided some evidence linking prenatal exposure to androgen excess with adverse programming of cardiovascular function in the offspring (16). Blood pressure measurement carried out over a 24-h period in conscious, undisturbed animals with the aid of radiotelemetry has revealed prenatal T-treated sheep to be hypertensive compared with control animals (59). A tendency for higher heart rate was also observed in the same study. At the cardiac level, prenatal T treatment increased the expression of several molecular markers involved in insulin signaling and those associated with cardiac hypertrophy and stress compared with control animals (60). In addition, myocardial disarray (multifocal) and increased cardiomyocyte diameter were observed in histological investigations (60). Together, these findings indicate that prenatal T excess results in adverse left ventricular remodeling in sheep, which likely contributes to the development of adult hypertension in these animals.

Effects of Prenatal Estradiol Excess

In contrast to the extensive information available on the impact of gestational T excess on cardiometabolic outcomes, studies investigating the effects of prenatal exposure to excess estradiol are scarce. Studies of the impact of gestational exposure to excess estradiol have focused primarily on reproductive (61, 62) and not metabolic outcomes. For the most part, information relative to prenatal estrogen programming of offspring metabolic function has been postulated from studies comparing the effects of prenatal T (aromatizable androgen) with prenatal DHT (nonaromatizable androgen) or with prenatal cotreatment with T and the androgen antagonist flutamide. Of the several organ systems affected by gestational

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exposure to T excess, the adipose tissue appears to be particularly vulnerable to prenatal estrogenic programming. For instance, our findings that prenatally blocking androgen action with an androgen antagonist (flutamide) or improving gestational insulin sensitivity with an insulin sensitizer (rosiglitazone) failed to prevent the increase in the ratio of small:large adipocytes suggest that the effects of prenatal T excess on adipocyte size distribution are likely mediated by the estrogenic effect of T. This possibility is supported by in vitro studies using adipocyte cell lines (63) and in vivo studies in transgenic mice (64) that reported inhibitory effects of estrogens on preadipocyte differentiation and adipocyte development. Although the different paradigms of prenatal androgen exposure (T, DHT, or T plus flutamide) suggest that some adult metabolic perturbations are programmed via the estrogenic pathway, future studies using prenatal estradiol and estradiol antagonist treatments are needed to confirm this premise.

Effects of Prenatal Glucocorticoid Excess

Exposure to glucocorticoid excess during prenatal life alters tissue differentiation and thus provides a common pathway by which environmental insults can program adult cardiometabolic disease (35, 65, 66). Fetomaternal concentrations of glucocorticoids are elevated under several conditions (maternal undernutrition, stress, hypoxia, and disease states) associated with the development of the adverse adult cardiometabolic phenotype in the offspring (65, 67, 68). Excess glucocorticoids inhibit fetal growth and disrupt the normal developmental trajectory of numerous metabolic organs, brain, heart, and the adrenal glands in the offspring (66). These adaptive changes have been linked to cardiometabolic, reproductive, and endocrine dysfunctions later in life (35, 65). Several animal models have been used to investigate the effects of intrauterine exposure to high levels of glucocorticoids on the offspring's health (65, 69–71). This review focuses primarily on effects of prenatal glucocorticoids on cardiometabolic function using sheep as the animal model. For more comprehensive reviews on the effects of early-life glucocorticoid exposure, readers are referred to work by Braun et al. (35), Fowden et al. (72), and Moisiadis & Matthews (68, 73).

Fetal and placental development

Experimental manipulation of fetal concentrations of glucocorticoids during late gestation by fetal adrenalectomy or infusion of exogenous cortisol has shown that cortisol promotes changes in tissue expression of cytostructural proteins, enzymes, receptors, and growth factors (72, 74). These modifications result in alterations in tissue morphology, metabolism, biochemical composition, and hormone sensitivity, with functional consequences for the offspring (72). For instance, chronic late-gestational infusion of hydrocortisone reduced placental weight by ~25%, fetal growth rate by ~30%, and spleen weight by 30% while increasing fetal heart weight, ventricular wall thickness, and fetal blood pressure (75) and inhibiting growth of the axial skeleton (76). Notably, fetal concentrations of IGF1 decreased during cortisol infusion, whereas IGF1-binding protein concentrations increased (75). In contrast, fetal adrenalectomy prevented the normal prepartum cortisol surge and increased the growth rate during late gestation. Late-gestational administration of low doses of the corticosteroid dexamethasone also decreased fetal body weight (77). However, the growth-suppressing effects of corticosteroid administration directly to fetal sheep were less severe

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than those seen with maternal administration (78), indicating that the growth-inhibiting effects of glucocorticoids are likely mediated, at least in part, via actions on the placenta (72, 79).

In support of placental involvement, administration of both natural and synthetic glucocorticoids during mid- to late-gestation has been shown to reduce placental weight (79), diminish expression of antiapoptotic markers, and elevate expression of proapoptotic factors in the placenta (72, 80). Gestational glucocorticoid excess was also found to increase the expression of glucocorticoid receptors in the placenta (77) and reduce more everted placentomes (81). Although the physiological significance of these modifications has not been completely elucidated, glucocorticoids have been shown to reduce glucose and amino acid transport across the placenta and alter its endocrine function (72, 79). For instance, glucocorticoid excess during early gestation reduces placental lactogen production, which can alter maternal metabolism and nutrient allocation to the gravid uterus (82). These findings suggest that glucocorticoid-induced changes in placental morphology, nutrient transport, and endocrine function appear to be critical mechanisms mediating the inhibitory effects of glucocorticoids on fetal growth in sheep.

Glucose-insulin homeostasis

Relative to the development of glucose-insulin homeostasis in the offspring, natural and synthetic glucocorticoids have different programming effects when administered during early pregnancy. For instance, cortisol but not dexamethasone induced basal hyperglycemia, whereas dexamethasone but not cortisol increased the first-phase insulin response to glucose in the adult male offspring (78). Similarly, cortisol, but not dexamethasone, administered between GD 26 and GD 28 resulted in fasting hyperglycemia and second-phase hyperinsulinemia during a glucose tolerance test in adult male offspring (83). In contrast, both glucocorticoids programmed hypertension in the adult offspring (84). These findings demonstrate that maternal glucocorticoid exposure during early pregnancy alters glucose homeostasis in the offspring in a glucocorticoid-specific manner.

Another important variable determining the metabolic phenotypic outcomes of prenatal glucocorticoid excess is the window and duration of exposure. For instance, the effects of multiple doses of betamethasone over a 14-day period in late pregnancy on glucose-insulin dynamics in the adult offspring differed from those of a single dose given during late gestation (85, 86). Whereas multiple doses of betamethasone increased both basal plasma insulin and insulin-to-glucose ratio in the offspring, a single dose of dexamethasone increased only the insulin-to-glucose ratio (86). Route of administration, whether at the maternal or fetal level, did not influence the outcome relative to glucose metabolism in the offspring (87), indicating direct fetal effects of glucocorticoids on this variable.

Another factor influencing the extent of adverse outcomes programmed by prenatal glucocorticoid excess is the postnatal age at which glucose-insulin homeostasis is assessed (72). Whereas some changes in postnatal growth and endocrine function are apparent soon after birth, metabolic perturbations become evident only later in adult life, barring cardiovascular abnormalities that are detected early on (72). Often, metabolic perturbations are detected in only one sex; for example, perturbed glucose-insulin homeostasis is seen in

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four- to five-year-old male but not female offspring exposed to dexamethasone during early gestation (83). Similar to what has been observed with the developmental trajectory of insulin sensitivity in prenatal T-treated sheep (Figure 2), a compensatory adaptation may explain the delayed manifestation of pathology in response to prenatal overexposure to glucocorticoids.

Although multigenerational studies are scarce in large animal models, findings in the ovine model provide some evidence that elevated glucocorticoids during pregnancy have the potential to program metabolic perturbations in a multigenerational manner. For instance, dexamethasone administration during late gestation abolished the neonatal leptin peak and increased circulating cortisol in F2 females (88). Moreover, during a feeding challenge, F2 offspring exhibited increased appetite, excessive weight gain and adiposity, and impaired insulin response to an intravenous glucose tolerance test.

Hypothalamic-pituitary-adrenal axis

Because of the central role that glucocorticoids play in regulating metabolic function, programming of hypothalamic-pituitary-adrenal (HPA) axis perturbations per se may lead to metabolic dysfunction in the offspring (72). Overexposure to glucocorticoids during fetal development can alter both basal and stimulated concentrations of cortisol postnatally and, hence, disrupt metabolic homeostasis. For instance, administration of betamethasone during the last third of pregnancy elevated basal cord blood concentrations of adrenocorticotropic hormone (ACTH) in the ovine fetus (89). Additionally, both prenatal maternal stress, a condition that increases glucocorticoid concentrations, and betamethasone administration increased the fetal cortisol responses to a hypotensive challenge, a physiological fetal stressor (90). A single late-gestational injection of betamethasone to the mother markedly increased the basal and stimulated cortisol concentrations in the offspring at one year of age (85). In addition, mid- to late-gestational dexamethasone treatment not only increased basal cortisol in F1 offspring but also increased ACTH and cortisol concentrations in F2 offspring, thus demonstrating multigenerational effects (91). Overall, although the effects of prenatal glucocorticoid excess on the HPA axis are dependent on the type of glucocorticoid used and timing of overexposure, the majority of studies in sheep show increased adult basal and stimulated concentrations of cortisol, indicative of postnatal HPA axis dysregulation (65).

At the tissue level, gestational administration of synthetic glucocorticoids reduced fetal brain weight at term (92). Prenatal glucocorticoid excess also reduced the hypothalamic expression of arginine vasopressin and corticotrophin-releasing hormone postnatally (93). At the pituitary level, late-gestational betamethasone injections reduced the mRNA expression of POMC, a precursor for ACTH synthesis, and diminished the expression of the enzymes involved in the posttranslational processing of POMC (93). At the adrenal level, betamethasone increased postnatal mRNA levels of ACTH receptor and 3β hydroxysteroid dehydrogenase and decreased steroidogenic acute regulatory protein and P450 17α hydroxylase (93). Collectively, these findings indicate that prenatal glucocorticoid excess results in molecular and cellular alterations at multiple levels of the HPA axis. Because glucocorticoids regulate glucose homeostasis and have largely opposing effects to those of

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insulin (94), perturbations in the HPA axis resulting in cortisol hypersecretion in adults likely play a role in mediating metabolic derangements.

Weight gain and adiposity

Limited observations in the ovine model suggest that prenatal exposure to excess glucocorticoids is also associated with alterations in weight gain, adiposity, and adipose tissue function. For instance, a single administration of dexamethasone at term increased body weight and fat mass in female but not male ovine offspring (95). Furthermore, maternal nutrient restriction during early to mid-gestation, a condition that increases stress response and glucocorticoid concentrations, increased adipose tissue expression of glucocorticoid receptor and induced obesity (96). As mentioned previously, prenatal dexamethasone treatment increased feed intake, body weight gain, and adiposity during a feeding challenge in F2 female offspring (88). Similarly, repeated betamethasone infusion during late gestation resulted in IUGR followed by early postnatal catch-up growth (87). Although studies in the ovine model are scarce, these findings suggest that prenatal overexposure to glucocorticoids may increase the risk of obesity and adiposity in the adult offspring. For additional information on the effects of prenatal glucocorticoids on adiposity and obesity in other animal models, readers are referred to work by Reynolds (97).

Cardiovascular function

Studies in the ovine model have linked fetal exposure to excess glucocorticoids with later occurrence of cardiovascular disorders, particularly hypertension. Administrations of both cortisol and dexamethasone at early pregnancy give rise to hypertension in the adult offspring (34, 84). However, the mechanisms by which hypertension is induced differ, with cortisol increasing peripheral vascular resistance but not cardiac output and dexamethasone yielding reverse outcomes in the adult offspring (23, 72). Notably, maternal administration of a single course of dexamethasone results in hypertension in the adult offspring when administered at GD 27 and GD 80, but not at GD 64, suggesting that cardiovascular outcomes depend on the gestational stage at which overexposure occurs (27, 29).

Periods of developmental adaptation may exist relative to the manifestation of cardiovascular dysfunction. For instance, hypertension is not observed in the neonatal lamb after prenatal dexamethasone treatment, although changes in the baroreflexes indicative of perturbations of the neural mechanisms controlling blood pressure are evident even at this early stage of life (30). However, hypertension is already evident at four months of age, becoming progressively more pronounced as animals approach adulthood (32). Therefore, studies in the ovine model suggest that glucocorticoid-programmed cardiovascular dysfunction is manifested earlier in life when compared with perturbations in glucose-insulin homeostasis, which are often not detected until adult life (72).

Although the mechanisms by which prenatal glucocorticoids program adult hypertension are not completely understood, alterations in renal development appear to play a role. Early gestational treatment with both cortisol and dexamethasone reduced fetal glomerular number at GD 140 and resulted in increased nephron glomerular filtration rate and hypertension during adulthood, implicating nephron deficit as a potential cause of hypertension in this

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ovine model (98, 99). Prenatal dexamethasone treatment during early gestation also increased the hypothalamic expression of angiotensin, a regulator of vasoconstriction (32), suggesting the involvement of a neuroendocrine mechanism in the prenatal programming of hypertension in this sheep model.

EFFECTS OF PRENATAL EXPOSURE TO ENVIRONMENTAL STEROID MIMICS ON CARDIOMETABOLIC FUNCTION

Effects of Bisphenol A

Environmental exposure to EDCs during critical periods of organ development has been postulated as an important contributor to the increased incidence of metabolic diseases in the last decades (100, 101). One such EDC is BPA, a carbon-based compound that is widely used in the manufacture of epoxy resins and polycarbonate plastics. Studies have reported the ubiquitous presence of BPA in water, air, and dust, as well as in most human fluids, including the maternal circulation and placental fluids (102–104). BPA has estrogenic and antiandrogenic activities and has the potential to program the development of obesity, insulin resistance, and cardiovascular diseases (101, 105).

Fetal growth

Prenatal BPA treatment from GD 30 to 90 (the same window used in studies with native steroids) resulted in lower birth weight without altering gestational length of female offspring (106). The effects on body weight were associated with differences in height and chest circumference of the newborn female lambs. Although changes in the IGF1 system appear to play a role in the IUGR seen in prenatal T-treated sheep (24, 31), circulating concentrations of free IGF1 or insulin were not altered in the prenatal BPA-treated offspring (106). In spite of this, as in prenatal T-treated sheep, growth rate tended to be higher between two and four months of age in prenatal BPA-treated sheep, indicative of compensatory growth during early postnatal development (106).

Glucose-insulin homeostasis

A dose-response study (0.05, 0.5, or 5 mg/kg/day) focusing on the offspring's metabolic health revealed that prenatal exposure to BPA at levels similar to those found in humans reduced insulin sensitivity in the female offspring during prepubertal age (107). The changes in glucose-insulin homeostasis were characterized by an increase in fasting glucose concentrations and a reduction in the insulin sensitivity index during an intravenous glucose tolerance test (107). During the postpubertal period (13 months of age), perturbations in insulin sensitivity were even more evident in these prenatal BPA-treated females, manifesting as increased mean cumulative insulin response and elevated cumulative insulin/glucose ratio (107). Acute insulin response to glucose administration was also higher in all BPA groups compared with control females. Although the longitudinal characterization of insulin sensitivity in this model is limited, these findings suggest that insulin resistance is established early in life (prepubertally) in prenatal BPA-treated sheep and deteriorates further as animals approach adulthood. In addition, prenatal BPA treatment negated the effects of postnatal overfeeding on insulin sensitivity, suggesting a complex interaction

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between prenatal BPA exposure and postnatal diet. These observations in the ovine model are of important translational relevance, considering that the BPA doses used in these studies yielded circulating levels similar to those reported in humans, and are corroborated by findings in rodent models (49) that also demonstrate that prenatal BPA exposure is associated with adverse metabolic outcomes.

Adiposity and oxidative stress

Perturbations in adipose tissue distribution, adipocyte morphometry, and adipose tissue inflammation and oxidative stress contribute to the development of adverse metabolic outcomes, such as obesity and insulin resistance, during adult life. Additionally, environmental insults occurring postnatally (second-hit) have the potential to exacerbate a preexisting condition caused by the first insult during prenatal development (first-hit). Thus, conditions such as postnatal excessive weight gain and/or exposure to EDCs can amplify or unmask adverse outcomes programmed prenatally (108, 109). When subjected to postnatal overfeeding, female sheep prenatally exposed to BPA manifested an increased ratio of visceral to subcutaneous fat compared with overfed control females, whereas no BPA-specific effects were observed in maintenance-fed animals (107). Given the association of visceral adipose tissue and negative metabolic outcomes (110), the extent to which increased visceral adiposity would ultimately amplify the severity of insulin resistance in prenatal BPA-treated, postnatally overfed animals is unclear. Similarly, although the adipocyte size distribution was shifted toward a larger area and diameter in the visceral adipose tissue of control overfed and BPA-treated sheep, highlighting the obesogenic potential of BPA (107), overfeeding did not exacerbate the effects of prenatal BPA on adipocyte size. In the subcutaneous adipose tissue, prenatal BPA exposure also increased the expression of CD68, suggesting increased macrophage infiltration (107). Collectively, these findings in the female sheep not only demonstrate a risk of metabolic perturbations during postnatal life following prenatal exposure to BPA at human-relevant levels but also highlight the complicated interaction that prevails between prenatal BPA and postnatal overfeeding/obesity.

Prenatal BPA treatment was also found to induce oxidative stress (111), a forerunner of adult metabolic disease (112). Using oxidized tyrosine moieties to study pathways involved in oxidative stress, Veiga-Lopez et al. (111) found that prenatal BPA treatment increases 3-nitrotyrosine (a marker for nitrosative stress) in GD-90 female fetuses, whereas a positive trend was observed at GD 65. Moreover, 3-nitrotyrosine was also elevated in both plasma and adipose tissue in prenatal BPA-treated sheep during adult life, demonstrating long-term effects of prenatal BPA exposure (111). This may also reflect the cumulative effects of BPA throughout the window of exposure (GD 30 to GD 90), because BPA has been found to accumulate in adipose tissue (113). Altogether, these observations in the ovine model demonstrating the induction of systemic and tissue-level nitrosative stress support a causal link between prenatal BPA and postnatal oxidative stress, because associations between BPA exposure and nitrosative stress have also been observed in human studies (111).

Cardiovascular function

Our studies addressing the interaction of prenatal BPA with postnatal overfeeding on the cardiovascular system in sheep found prenatal BPA had no effect on blood pressure or

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morphometric measures but increased the expression of atrial natriuretic peptide, a marker of left-ventricular hypertrophy frequently associated with systemic hypertension, suggestive of the impending risk of adverse left-ventricular programming by BPA (114). Moreover, prenatal BPA exposure reduced collagen expression in the right ventricle, which could result in altered systolic function (114). Interestingly, whereas postnatal overfeeding resulted in a marked increase in diastolic blood pressure, prenatal BPA treatment prevented the increase in blood pressure and increase in left-ventricular surface area caused by overfeeding, indicative of complex interactions in the manifestation of prenatally programmed events relative to the postnatal environment (114). Collectively, these observations suggest that prenatal exposure to BPA alone, while having no evident effect on cardiac structure, may induce changes at the molecular level that could reduce compliance and prevent compensatory responses to postnatal overfeeding in sheep (114).

Assessment of transcriptional changes by RNA-Seq analysis of the myocardium to better understand the paradoxical interaction between prenatal BPA and postnatal overfeeding revealed transcriptional dysregulations in both the prenatal BPA and postnatal overfed groups. Although the biological pathways they affected were remarkably similar, the genes that were affected in these two groups differed considerably (115). Furthermore, similar to the effects on cardiovascular function, the combined treatment failed to produce a synergistic response; on the contrary, prenatal BPA exposure partially prevented the effects of postnatal overfeeding (115). Additional studies addressing whether these unexpected observations are adaptive or maladaptive are needed to help us better understand the interactive effects of prenatal BPA and postnatal obesity on adult cardiovascular function.

Effects of other environmental steroid mimics

Public pressure has led to a shift away from BPA in several consumer products, resulting in a marked increase in the use of other bisphenolic chemicals, such as bisphenol S (BPS) (116), another environmental steroid mimic recently demonstrated to act as an obesogen (117). Of concern, BPS has been detected in 70% of human urine samples (118, 119) and fetal cord blood (120). Recent studies in sheep have shown that mid-gestation (GD 30–100) exposure to BPS reduced maternal levels of pregnancy-associated glycoproteins (121) secreted by the trophoblast layer of the placenta and considered to be biomarkers of pregnancy health in ruminants (122). Gestational BPS treatment also reduced the maternal progesterone, supportive of impaired placental endocrine function (121). Importantly, BPS was shown to cross the ovine placental barrier and reach the fetus in high concentrations (121). At the fetal level, gestational BPS treatment altered gene expression of steroid receptors in preadipocytes and impaired terminal differentiation of adipocytes in male but not female fetuses (123), indicative of sex-specific effects of gestational BPS exposure. Although these findings demonstrate important effects of BPS on placental and fetal development, in the absence of information on the postnatal repercussions of gestational exposure to BPS, a key requirement for determining if these adaptive changes culminate in adult pathology, the adult metabolic risks associated with these early developmental changes induced by prenatal BPS exposure remain unclear.

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Methoxychlor (MXC) is a synthetic organochlorine with estrogenic and antiandrogenic activity that was used as a pesticide in the United States until 2003. Although its use has been banned for over a decade, MXC is a persistent chemical that still exists in the environment and in the circulation of men and women (124). Mid-gestational MXC treatment resulted in high concentrations in the fetus, indicating it crosses the placental barrier (106). Although it did not impair fetal growth, gestational MXC treatment had long-term effects on several reproductive endocrine parameters in the female sheep (106). Considering MXC impairs metabolic function in rodents (125), it will be important in the future to characterize the metabolic outcomes associated with gestational MXC exposure in the ovine model.

CONCLUSIONS

Potential Mechanisms

Although prenatal exposure to excess androgens, glucocorticoids, and EDCs with steroidogenic activity can disrupt the developmental trajectory of the fetus, culminating in adult disease, there are multiple similarities in the outcomes programmed by these different prenatal insults (Table 1). For example, changes in placental function and reduced fetal growth are common consequences of prenatal T (32, 36) and glucocorticoid excess in sheep (75). Moreover, cardiometabolic derangements, such as insulin resistance, increased adiposity, and hypertension, are mutual phenotypic traits programmed by prenatal exposure to different steroid hormones and steroid mimics. The similarity in these phenotypic outcomes suggests that common mechanisms may be involved.

Depending on the steroid hormone/mimic studied, organ system affected, and window of exposure, several mechanisms have been proposed. It is becoming increasingly evident that distinct prenatal insults can induce fetomaternal responses that converge on mutual mechanistic pathways to reprogram the adult phenotype of the offspring (Figure 3) (3). Steroid hormones such as androgens, estrogens, and glucocorticoids, as well as metabolic hormones, such as insulin and leptin, regulate several key physiological processes during fetal development and are potential targets of perinatal insults (126–130). Animal studies have shown that different prenatal insults, such as maternal undernutrition (2, 131), a condition that increases glucocorticoid levels, as well as gestational exposure to T excess (25), increase maternal insulin and leptin concentrations. Moreover, both gestational undernutrition and stress increase not only maternal glucocorticoids but also maternal androgens (132–134). Similarly, prenatal T treatment increases not only androgen levels in the fetomaternal circulation but also maternal insulin and fetal estrogen levels (25). Studies employing cotreatment with T and androgen antagonists or insulin sensitizers suggest that the programming effects of T are mediated via the androgen receptor and insulin signaling, pointing to the potential involvement of other pathways, such as conversion of T into estradiol and activation of estrogen receptors. Findings that prenatal exposure to BPA, an EDC with estrogenic activity, replicates several phenotypic alterations seen in prenatal T-treated sheep provide evidence for fetal programming via estrogenic pathways. Therefore, endocrine alterations (e.g., hyperandrogenism and hyperinsulinemia) during gestation may

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serve as common mechanisms by which different prenatal insults can program similar adverse adult outcomes in the offspring.

Epigenetic modifications, such as DNA methylation, histone modification, and differential expression of noncoding RNAs, are emerging as important mechanisms of fetal programming (135). Because epigenetic modifications are inherently plastic and can accumulate over time, prenatal insults can have marked effects on this process, resulting in altered expression of several genes controlling the normal trajectory of fetal development (136, 137). For instance, prenatal dexamethasone treatment alters the DNA methylome of the hypothalamus, liver, and adrenal glands in rodents (138–140). Similarly, prenatal T excess alters the DNA methylation patterns in both infant and adult visceral adipose tissues in female nonhuman primates (141). Notably, epigenetic alterations appear to be important not only in programming the phenotype of the offspring but also in mediating transgenerational inheritance of traits (142).

Oxidative stress, a deleterious process that leads to damage of proteins, lipids, and DNA, is another important mechanism believed to underlie fetal programming (143). Gestational insults associated with IUGR, low birth weight, and adult disease, such as prenatal T and glucocorticoid excess, increase oxidative stress (47, 144). Prenatal BPA treatment, similar to native steroids, impairs fetal growth and promotes systemic and tissue-specific oxidative stress (111). Moreover, IUGR has been shown to promote low-grade inflammation in the adult offspring, which may serve as a potential mediator linking reduced fetal growth and the development of chronic disease (145). Therefore, both oxidative stress and inflammation have been postulated as important mechanisms underlying adverse cardiometabolic outcomes associated with prenatal steroid excess and EDC exposures (146). The observation that maternal antioxidant supplementation decreases oxidative stress and improves metabolic function in the offspring of a rodent model of gestational obesity further indicates that inflammation and oxidative stress are likely contributors to the programming of adverse metabolic outcomes (147).

Finally, early postnatal perturbations resulting from in utero exposure to steroid excess, such as endocrine imbalances and metabolic disruptions, themselves may play a role in unmasking and/or amplifying the defects programmed prenatally. For instance, postnatal excessive weight gain exacerbates the metabolic derangements programmed by prenatal T excess (38). Furthermore, postnatal treatment with an insulin sensitizer not only restores insulin sensitivity but also improves other metabolic parameters in adult sheep prenatally exposed to T excess (47), suggesting that postnatal imbalances in glucose-insulin homeostasis are required for the full manifestation of the disease phenotype. Thus, it is apparent that although prenatal exposure to steroid excess has important organizational effects, postnatal perturbations in endocrine and metabolic function have activational effects impacting the development, manifestation, and/or severity of the disease phenotype in the offspring (3, 14).

Future Directions

Epidemiological data and studies in animal models, including the ones in the sheep models discussed above, demonstrate that in utero exposure to steroid excess or steroid mimics can

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impact the offspring's cardiometabolic health. However, the scarce information on the effects of such insults on the fetomaternal environment and limited longitudinal phenotyping studies hinder our understanding of the windows of susceptibility and the potential short- and long-term effects on relevant organ systems. Moreover, the interaction between prenatal steroid excess and resulting early postnatal perturbations on the development of cardiometabolic diseases remains poorly understood. Thus, the sheep model represents a valuable tool to investigate the long-term effects of prenatal exposure to native steroids and EDCs on the offspring's health, the cellular and molecular mechanisms involved, and testing of novel intervention strategies.

Animal models with a developmental trajectory similar to humans, such as sheep, are also important research tools to investigate the transmission of phenotypic traits to subsequent generations. Studies in different animal models have shown that prenatal exposure to steroid excess results in multigenerational modifications in the epigenome and cardiometabolic phenotype (91, 148–150). Future studies should focus on distinguishing the role of epigenetic inheritance from the vertical transmission of endocrine perturbations, such as hyperandrogenism and hyperinsulinemia, which may facilitate perpetuation of pathologic outcomes to subsequent generations.

Because prenatal steroid excess can impact multiple physiological systems, future research should also gain a better understanding of the interplay between these systems in the development of metabolic dysfunctions. For example, prenatal T excess impacts both the reproductive and metabolic systems, and reproductive perturbations (e.g., functional hyperandrogenism) impair metabolic function, whereas metabolic imbalances (e.g., insulin resistance and hyperinsulinemia) can impact reproductive function, thus forming a vicious cycle (3). Consequently, interventions targeting multiple organ systems may be needed to prevent the manifestation of adverse outcomes programmed prenatally. Observations in women with PCOS support this premise, in which combined treatment with an androgen antagonist and an insulin sensitizer yields additive benefits in improving metabolic function compared with monotherapies (151). Finally, the identification of common mediators activated by prenatal exposure to different steroids and/or EDCs in models such as sheep can help identify early biomarkers and potential therapeutic targets for intervention.

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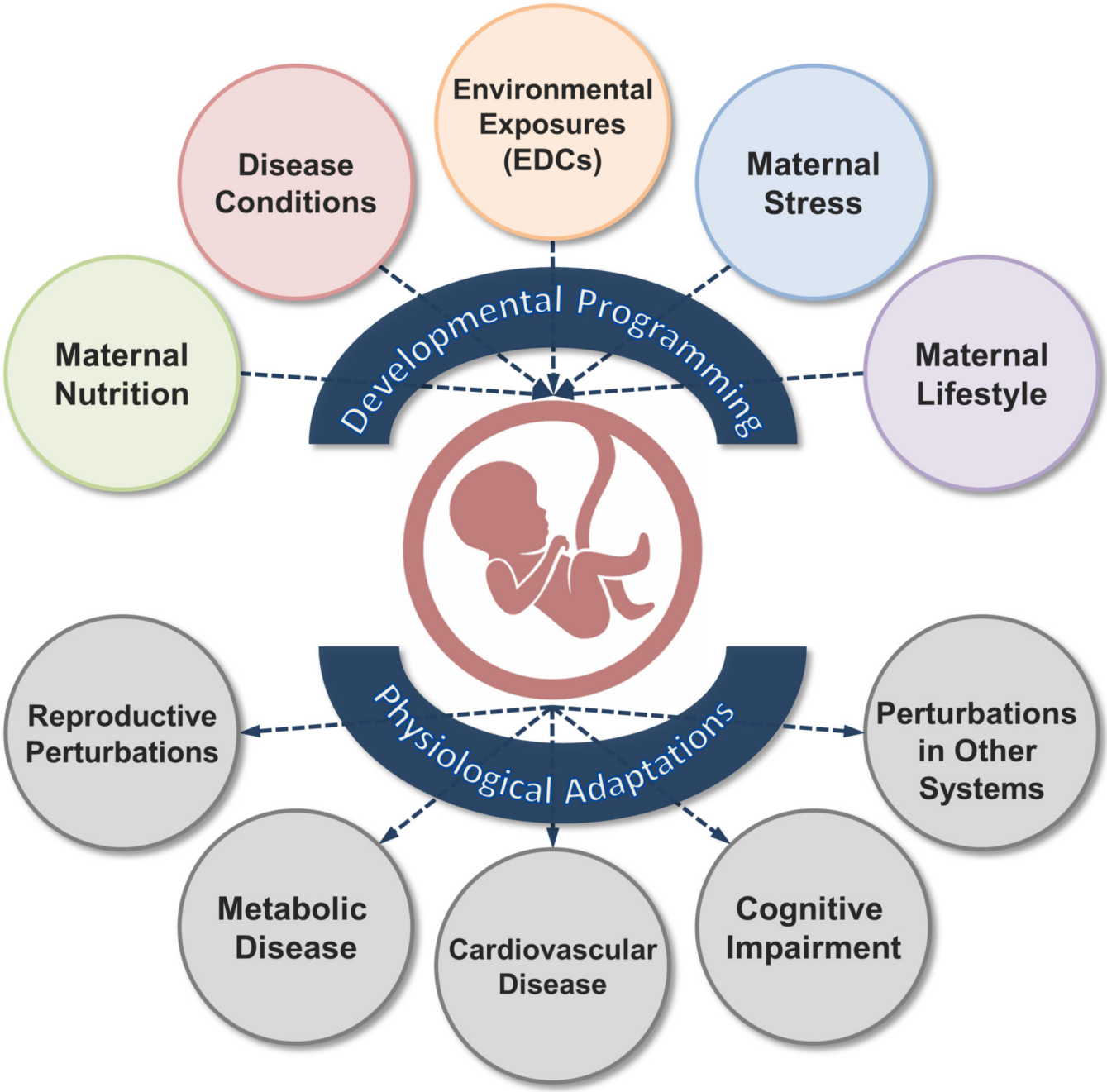

Figure 1.

Schematic illustrating the effects of prenatal insults in programming adverse health outcomes in the offspring. Prenatal exposure to different insults during critical periods of fetal development results in physiological adaptations that often prove to be detrimental and result in failure to overcome programmed pathology. Abbreviation: EDC, endocrine-disrupting chemical.

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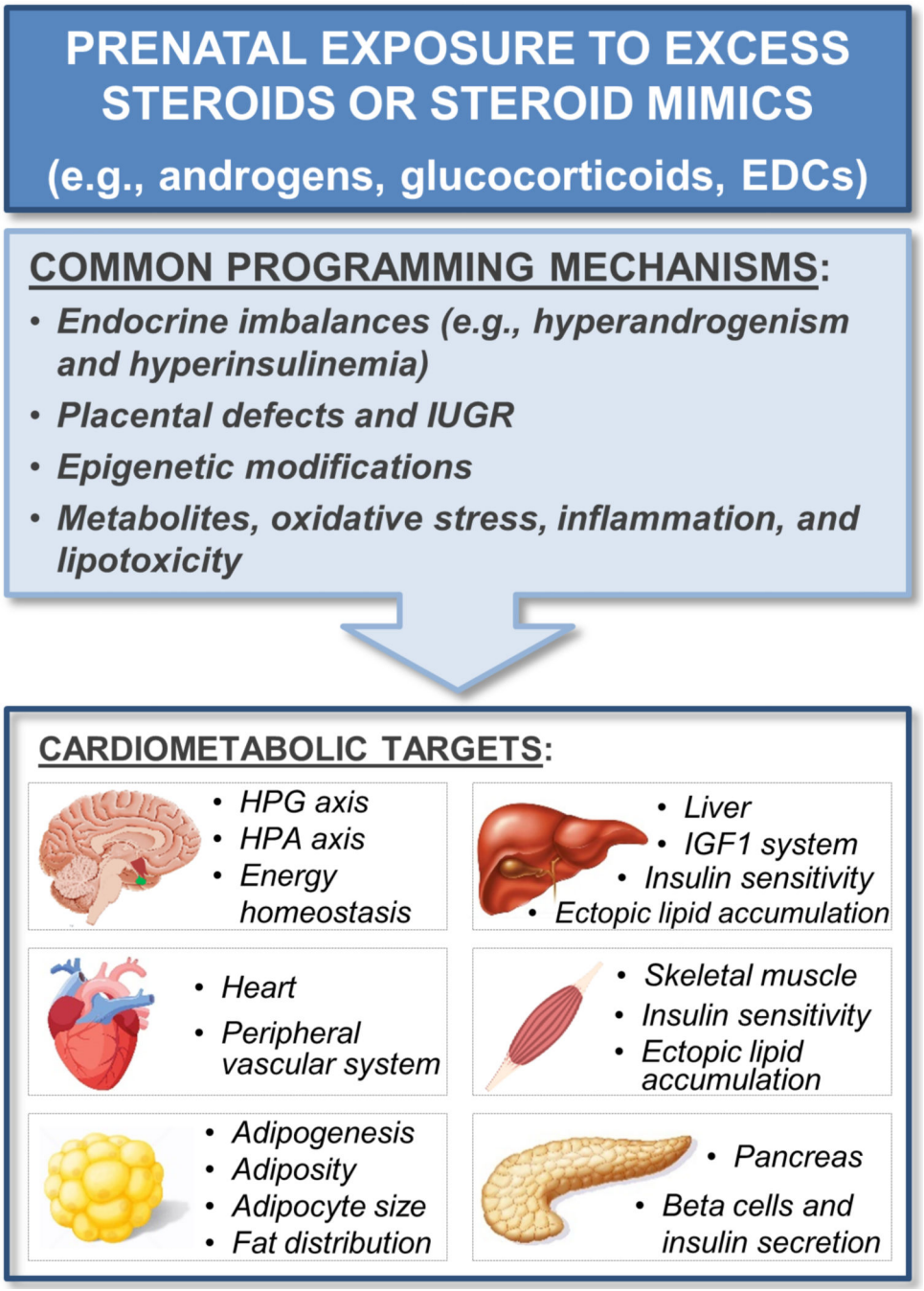

Figure 2.

Schematic model for developmental changes in insulin sensitivity in female sheep prenatally exposed to testosterone (T) excess. This working model is supported by findings of insulin resistance during early life in prenatal T-treated sheep, followed by elevated insulin sensitivity at postpubertal age, normal insulin sensitivity during early adulthood, and reestablishment of insulin resistance later in adult life. Collectively, observations in the ovine model are suggestive of the existence of a period of compensatory adaptation of the metabolic tissues to prenatal exposure to excess T.

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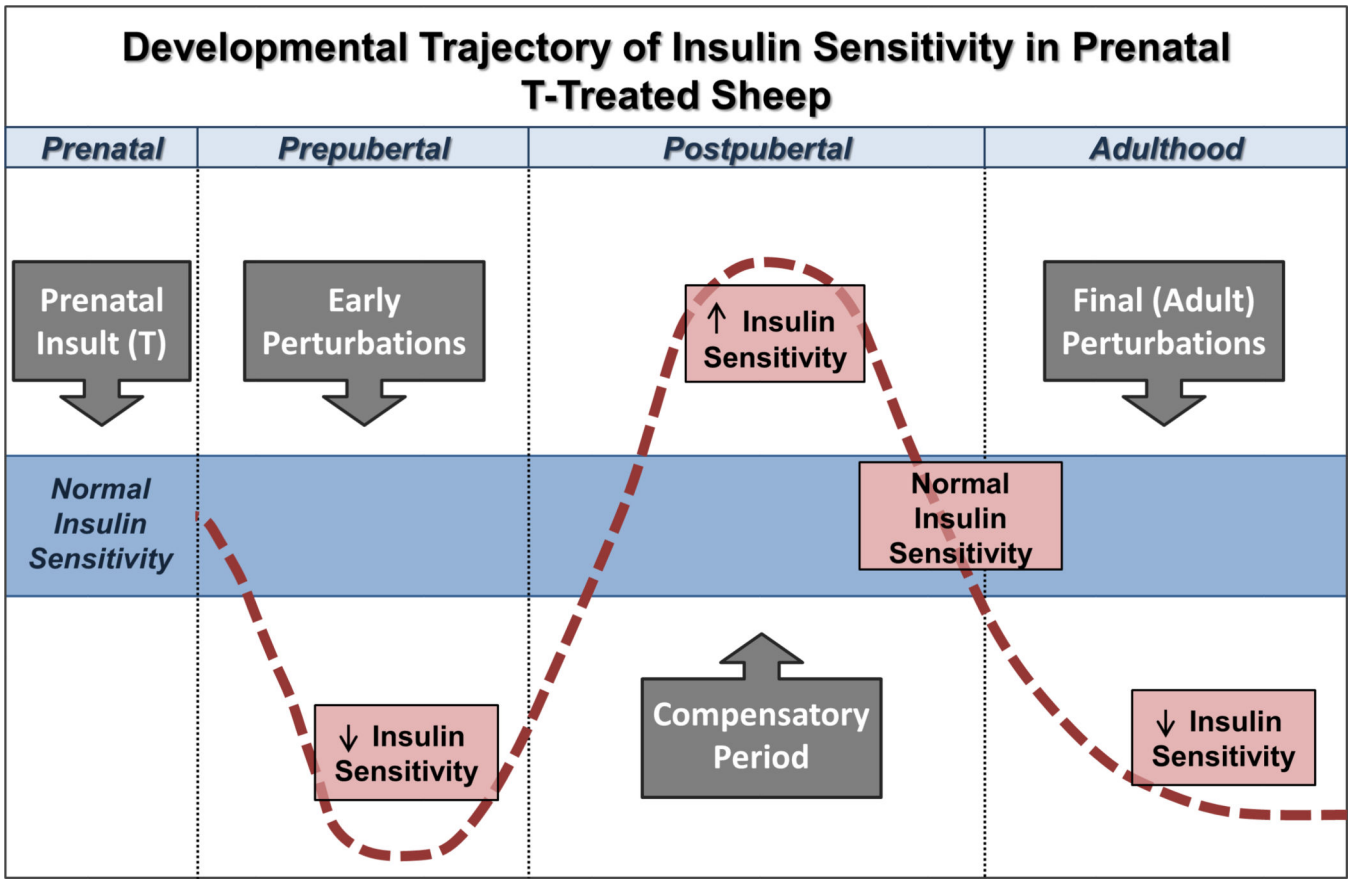

Figure 3.

Schematic illustrating the effects of prenatal exposure to steroid excess on cardiometabolic organ systems. Prenatal exposure to native steroids (e.g., androgens and glucocorticoids) or endocrine-disrupting chemicals (EDCs) with steroidogenic potential (e.g., bisphenol A) act via common mechanistic pathways to reprogram several organ systems in the offspring resulting in the development of adult cardiometabolic disease. Abbreviation: HPA, hypothalamic-pituitary-adrenal; HPG, hypothalamic-pituitary-gonadal; IGF1, insulin-like growth factor 1; IUGR, intrauterine growth restriction. Source of images used in this figure: https://pixabay.com..

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Table 1

Summary of cardiometabolic characteristics of the different ovine models of prenatal exposure

Phenotypic parameters	Prenatal exposure to testosterone	Prenatal exposure to glucocorticoids	Prenatal exposure to BPA
Fetomaternal endocrine milieu	Maternal hyperandrogenism Maternal hyperinsulinemia Reduced maternal progesterone Reduced maternal medium-chain acylcarnitines Fetal hyperandrogenism Elevated fetal estradiol Reduced fetal IGF1	Increased maternal levels of glucocorticoids Increased fetal levels of glucocorticoids Reduced fetal levels of IGF1 and increased IGF1 binding proteins Elevated fetal concentrations of ACTH	Elevated levels of BPA in the fetal circulation Normal fetal levels of IGF1 and insulin
Fetal development	Late gestation IUGR Reduced liver, kidney, and spleen	IUGR Increased fetal heart weight Reduced fetal spleen weight	IUGR
Placental development	Advanced placental differentiation	Reduced placental weight Increased placental expression of glucocorticoid receptors Reduced placental lactogen	Not studied
Glucose-insulin homeostasis	Insulin resistance Compensatory hyperinsulinemia Liver: insulin resistant Muscle: insulin resistant Adipose tissue: insulin sensitive Temporal fluctuations in insulin sensitivity during postnatal life	Hyperglycemia Insulin resistance Sexually dimorphic effects on glucose-insulin homeostasis Multigenerational effects (insulin resistance in F2 females)	Hyperglycemia Insulin resistance Increased acute insulin response to glucose administration
Hypothalamic neurocircuitries	Increased number of AgRP neurons in the arcuate nucleus Increased AgRP fiber density in several hypothalamic regions	Reduced fetal brain weight Reduced hypothalamic expression of arginine vasopressin Reduced hypothalamic CRH	Not studied
Adipose tissue	Increased ratio of small to large adipocytes Ectopic lipid accumulation Elevated saturated free fatty acids Oxidative stress	Increased weight gain and adiposity in females Increased expression of glucocorticoid receptors in the adipose tissue	Increased ratio of visceral to subcutaneous fat Larger adipocytes in the visceral adipose depot Increased CD68 expression in the subcutaneous adipose tissue Oxidative stress
Cardiovascular function	Hypertension Histological alterations in the left ventricle	Hypertension Nephron deficit	Increased ANP expression in the left ventricle Reduced collagen expression in the right ventricle

Abbreviations: ACTH, adrenocorticotropic hormone; AgRP, Agouti-related peptide; ANP, atrial natriuretic peptide; BPA, bisphenol A; CD68, cluster of differentiation 68; CRH, corticotropin-releasing hormone; IGF1, insulin-like growth factor 1; IUGR, intrauterine growth restriction; T, testosterone.